National University of Computer and Emerging Sciences



Laboratory Manual # 06
Data analysis and visualization

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Objectives:

- Introduction to Power BI and its interface
- Importing Data
- Data Cleaning with Power Query
- Data Modeling and Relationships
- Creating Visualizations

Dataset for this lab: **Retail_Sales_Data.xlsx** (provided with manual)
Submission: You have to submit a word document (containing screenshots and answers) and your pbix file

Introduction:

Power BI is Microsoft's business-intelligence tool. It connects to data (Excel files, databases, web services), lets you clean and organize that data, calculate business figures, and present the results as interactive charts.

Term	Meaning
Data model (dataset)	The tables you loaded, the relationships between them, and the calculations (measures) you defined. Visuals read from the model.
Visualization (visual)	One chart, card, table or slicer on a page.
Report	A file made of one or more pages. Each page holds visuals. This is what you build in Desktop.
Dashboard	In the Power BI Service, a single page of tiles is pinned from reports. In everyday language (and in this lab) "dashboard" also means a well-designed single report page that summarizes key figures.
Field	A column (or measure) in your data that you can drag into a visual.
Power Query	The built-in tool for cleaning and reshaping data before it is loaded into the model.
DAX	Data Analysis Expressions: the formula language used to create measures and calculated columns.

Every Power BI project follows the same six steps.

Get Data	→	**Transform Data**	→	**Model Data**	→	**Create Measures**	→	**Visualize**	→	**Analyze**

Power BI interface:

Area	What it is used for
Ribbon	The tabs along the top (File, Home, Insert, Modeling, View, Help …). Contains the commands: Get data, Transform data, New measure, New table, etc. Some tabs (Table tools, Column tools, Measure tools) appear only when you select a table, column or measure.
Canvas	The large white page in the middle where you place and arrange visuals.
Report view	Top icon on the left-hand side (bar-chart icon). Where you build pages and visuals.
Table view (Data view)	Second icon (grid icon). Shows the actual rows of the table you select so you can inspect values, data types and formats.
Model view	Third icon (three connected boxes). Shows tables as boxes and relationships as lines between them.
Data pane (Fields)	Far right. Lists every table and field in your model. You drag fields from here into visuals.
Visualizations pane	Right side. The gallery of visual types (top), the field "wells" such as X-axis, Y-axis, Legend (middle), and the **Format visual** button (paintbrush icon) for colours, titles and labels.
Filters pane	Narrow pane between the canvas and the Visualizations pane. Shows filters that apply to the selected visual, the page, or all pages.
Formula bar (DAX area)	A bar that appears under the ribbon when you create a measure, calculated column or calculated table. You type DAX formulas here.

Dataset:

You will use **Retail_Sales_Data.xlsx**. It is an extract from a retail company's sales system covering orders placed between **4 January 2025 and 31 December 2025**. The workbook has one sheet, **Sales**, in which each row is one order line. The data was exported directly from the source system, so it may not be perfectly clean: finding and fixing any problems is part of the lab.

Column	Meaning	Role
OrderID	Unique order reference (ORD001 …)	Identifier
OrderDate	Date the order was placed	Dimension (date)
CustomerName	Name of the customer business	Dimension
Region	Sales region: North, South, East, West	Dimension
CustomerSegment	Consumer, Corporate or Small Business	Dimension
Product	Product sold (12 different products)	Dimension
Category	Product category (4 categories)	Dimension

Quantity	Number of units sold	Measure (numeric)
UnitPrice	List price of one unit before discount	Measure (numeric)
Discount	Discount given, as a fraction (0.05 = 5%)	Measure (numeric)
Sales	Revenue of the order after discount: Quantity × UnitPrice × (1 − Discount)	Measure (numeric)
Profit	Profit made on the order	Measure (numeric)
PaymentMethod	Credit Card, Bank Transfer, Mobile Wallet or Cash	Dimension
InternalNotes	Free-text notes written by staff	Not needed for analysis

Dimensions and measures

- A **dimension** (categorical field) describes *what* or *who* or *when*: Region, Category, Product, Customer Segment, Date. You use dimensions to group and filter.
- A **measure** (numeric field) is a number that can be added up or averaged: Sales, Profit, Quantity. You analyse measures *by* dimensions (for example, “Sales **by** Category”).

The tables you will have at the end of Task 3

The raw file is a single “flat” table. Professional Power BI models separate descriptive information into small lookup tables. During the lab you will create the following model:

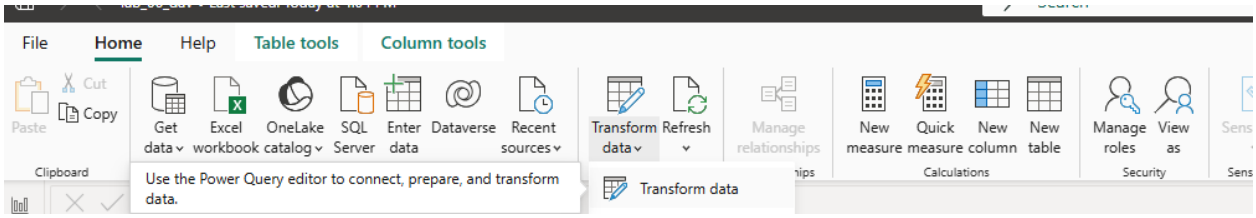
Table	Type	Contents	Key column
Sales	Fact table	One row per order (200 rows after cleaning): dates, quantities, Sales, Profit, etc.	OrderID (unique); foreign keys: OrderDate, Product, Region
Dim_Product	Dimension table	One row per product with its category (12 rows)	Product (unique)
Dim_Region	Dimension table	One row per region (4 rows)	Region (unique)
Dim_Date	Dimension table	One row per calendar day of 2025 with Month, Quarter, etc. (365 rows)	Date (unique)

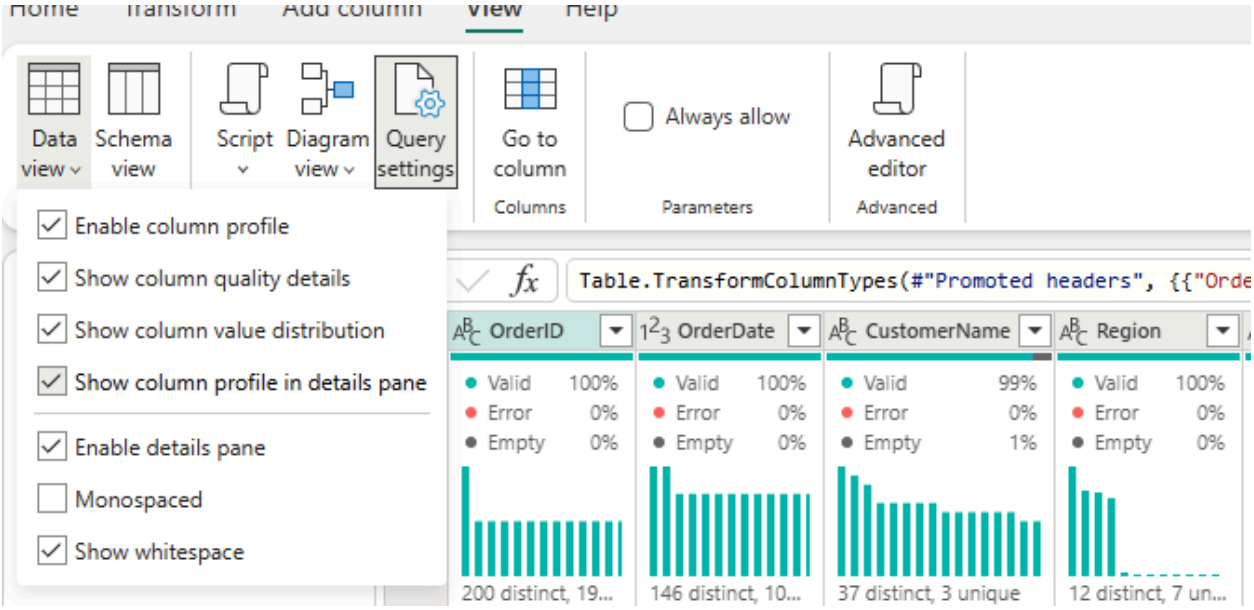
A **fact table** records events (here: sales) and contains numbers to analyse. A **dimension table** lists the “things” the events relate to. Relationships link a key in a dimension table to the matching column in the fact table.

Task 1: Importing data

[Note: When Power BI imports data it reads each column and assigns a **data type** (Text, Whole number, Decimal number, Date, etc.). The type controls what you can do with a column (you can only sum numbers, only use dates in a time axis). Wrong data types are one of the most common problems.]

1. Open Power BI Desktop and select Blank report. Immediately save with File → Save as using the name Lab_Retail_<YourStudentID>.pbix.
2. On the **Home** tab, click **Get data**. A list of common sources appears. Select **Excel workbook**. (If you do not see it, choose **More…**, then **File → Excel workbook → Connect**.)
3. Browse to **Retail_Sales_Data.xlsx** and click **Open**.
4. The **Navigator** window opens. Tick the box next to **Sales**. A preview of the data appears on the right.
5. Click **Load** (not "Transform Data" this time). Power BI loads the table into the model.
6. In the **Data pane** (right), click the arrow beside **Sales** to expand it. You now see every column as a field.
7. Click the **Table view** icon (second icon, left side). The rows appear. At the bottom-left of the window, Power BI shows the table name and number of rows. **Record the number of rows.**
8. Click the header of the **Quantity** column. On the ribbon, the **Column tools** tab shows the column's **Data type**. Check the data type of each column and complete the table below.
9. Scroll through the data. Look for blank cells, strange spellings, repeated rows, and columns you may not need.
10. Now open the profiling tools: on the **Home** tab, click **Transform data** (this opens the **Power Query Editor** in a new window). On the **View** tab, tick **Column quality**, **Column distribution** and **Column profile**. These show, above each column, the percentage of valid, error and empty values and the number of distinct values. If the bottom bar says "Column profiling based on top 1000 rows", click it and choose **Column profiling based on entire data set**.





Take a screenshot of the results.

#	Question	Your answer
1.1	How many rows does the Sales table contain?	
1.2	How many columns does it contain?	
1.3	List the columns that contain **numerical** data (as intended by the business).	
1.4	List the columns that contain **categorical** data.	
1.5	Which column represents the date? What data type did Power BI assign?	
1.6	What data type did Power BI assign to **Quantity**? Is that sensible? Why or why not?	
1.7	Which column(s) contain missing (empty) values? How many missing values are there?	
1.8	Look at the **Region** column. How many distinct values does the Power Query profile report? How many *should* there be?	
1.9	List every data-quality problem you can find (missing values, inconsistent spelling, duplicates, wrong types, unnecessary columns).	

Expected: you should have found multiple issues within the dataset.

Task 2: Data cleaning with Power Query

Power Query is the data-preparation tool inside Power BI. You work on a copy of the data: every action you take is recorded as a **step** in the **Applied Steps** list (right side of the editor), and the original Excel file is never changed. Steps run in order each time the data is refreshed, so the cleaning is repeatable. You can click any step to see the data at that point, or click the ✕ beside a step to undo it.

1. Rename Columns

Double-click a column header, type the new name, and press **Enter**. (or right click on column name and select rename, after changing press enter)

- Rename PaymentMethod to Payment Method.
- Rename CustomerSegment to Customer Segment.

2. Change data types

The small icon at the left of each column header shows its type (ABC = text, 123 = whole number, 1.2 = decimal, calendar icon = date). Click the icon to change the type.

- Quantity (ABC): click the icon and select Whole Number. If Power BI asks whether to replace the current step, choose Replace current.
- OrderDate: if it shows a date-and-time icon, set it to Date.
- UnitPrice, Sales, Profit: set to Fixed decimal number (money-style, 4 decimals stored).
- Discount: set to Decimal Number.
- Leave OrderID, CustomerName, Region, Customer Segment, Product, Category and Payment Method as Text.

[Note: If a type change shows “Error” in any cell, there is a value that cannot be converted. Click the step before and look for stray text or spaces. Do not continue until the column shows 0% Error.]

3. Remove unnecessary columns

InternalNotes contains staff comments that are not part of the analysis.

- Click the InternalNotes column header to select it. Right-click and choose Remove (or Home → Remove columns → Remove columns).

4. Remove duplicate records

Because **OrderID** must be unique, any repeated OrderID is an error that would be double-counted in the totals.

- Click the OrderID column header to select only that column.
- Check its statistics for number of rows, count, distinct count etc
- On the Home tab choose Remove Rows → Remove Duplicates.
- Check the statistics again and record the new number of rows.

5. Handle missing values

Empty (null) values in a text column appear in charts as “(Blank)”. For CustomerName we will replace blanks with the label **Unknown**, so those orders still count in the totals but are clearly identified.

- Select the CustomerName column.
- On the Transform tab, click Replace Values.
- In Value To Find type null (all lowercase). In Replace With type Unknown. Click OK.
- Open the column’s filter drop-down (▼) and confirm that Unknown now appears and there are no empty entries.

6. Standardised text

Computers treat ‘east’, ‘East’, ‘EAST’ and ‘East ’ (with a trailing space) as different values. Two transformations fix this: Trim removes leading/trailing spaces and Capitalize Each Word makes every value look like ‘East’.

- Select the Region column.
- On the Transform tab click Format → Trim.
- With Region still selected click Format → Capitalize Each Word.
- Look at the column distribution / filter list: only four regions should remain.

7. Filter invalid records

Business rules tell us what valid data looks like: you cannot sell zero or negative units, and revenue cannot be zero or negative. Filtering is a safety net that removes rows breaking these rules.

- Click the drop-down arrow (▼) on the Sales column header → Number Filters → Greater Than… → enter 0 → OK.
- Do the same for Quantity (Greater Than 0).

8. Add a derived column

You will create a **Discount Band** column that groups orders into three simple categories, which is easier to analyse than five separate discount percentages.

- Go to the Add Column tab and click Conditional Column.
- Set New column name to Discount Band.
- Clause 1: If Discount equals 0 Then No Discount.
- Click Add clause. Clause 2: Else If Discount is less than or equal to 0.1 Then Low Discount.
- In else type High Discount. Click OK.
- Confirm the new column’s data type is Text.

[Take screenshot of applied steps]

9. Create the Product and Region lookup tables

To build a proper star-schema model (Task 3), create small dimension tables. Power Query lets you create a new query that references (re-uses) the cleaned Sales query.

- In the Queries pane, right-click Sales and choose Reference. A new query called Sales (2) appears.
- Rename it: right-click → Rename → Dim_Product.
- With Dim_Product selected, hold Ctrl and click the Product and Category column headers. Right-click and choose Remove Other Columns.
- Click the Product column header only, then Home → Remove Rows → Remove Duplicates. Check the row count: this is the number of distinct products.
- Right-click Sales again → Reference → rename the new query Dim_Region.
- Select the Region column → right-click → Remove Other Columns, then Remove Duplicates. You should have one row per region.

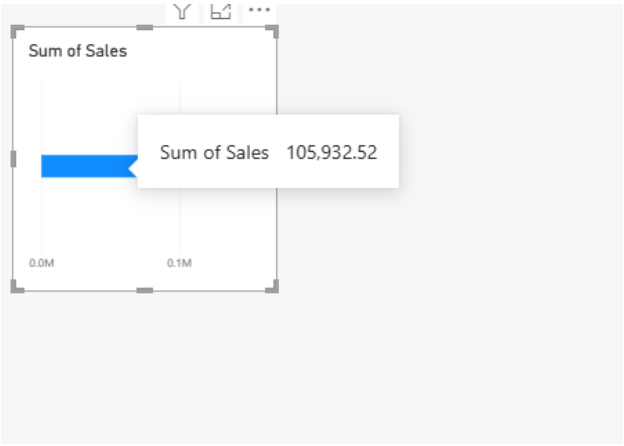
[Note: Do not delete the Product, Category or Region columns from the **Sales** query afterwards. The lookup tables depend on it.]

10. Close and apply

- Click Home → Close & Apply (top-left). Power BI loads the cleaned tables into the model.
- Save your report (Ctrl + S).

#	Question	Your answer
2.1	How many rows did the Sales table have before and after removing duplicates? Which OrderID was duplicated?	
2.2	Which Region values did you have to fix? List three examples of the original wrong spellings.	
2.3	How many CustomerName values were missing, and what did you replace them with?	
2.4	Did the “Greater than 0” filters remove any rows? Why is it still a good idea to include them?	
2.5	How many rows are in Dim_Product and Dim_Region?	
2.6	In **Table view**, select Sales. How many rows and columns does it now have?	

Verification:
In the Report view, add a temporary Card visual and drag Sales[Sales] into it (it will show “Sum of Sales”). The value must be 105,932.52. If it is different, your cleaning is incomplete. Delete this card afterwards. [You can check Task 5 **General method for every visual**]



Task 4: Data Modeling and relationships

A **data model** is the set of tables in your report and the **relationships** that connect them. A **relationship** tells Power BI: “the Product column in table A matches the Product column in table B”. Once a relationship exists, selecting a category on a slicer automatically filters the sales rows that belong to that category.

Key concepts

Concept	Explanation
Primary (unique) key	A column whose values never repeat in its table (OrderID in Sales; Product in Dim_Product; Region in Dim_Region; Date in Dim_Date).
Foreign key	A column in the fact table that holds a key from a dimension table (Sales[Product], Sales[Region], Sales[OrderDate]).
One-to-many (1 to many)	One row in the dimension table matches many rows in the fact table (one product → many sales rows). The “one” side is always the dimension table.
Filter direction	Filters flow from the “one” side to the “many” side (Dim_Product → Sales) so choosing a Category filters Sales.
Fact vs dimension table	Fact = events with numbers to analyse (Sales). Dimension = descriptive lists you filter and group by (Product, Region, Date). This layout is called a **star schema**.

1. Create a Date table

Analysing by month needs a continuous calendar. You will create one with DAX (you will study DAX later; for now just type the formula exactly as shown). A **calculated table** is a table created by a DAX formula rather than loaded from a file.

- Open Report view. On the Modeling tab, click New table. The formula bar appears.
- Delete any text in the formula bar and type or paste the formula below. Press Enter.

```
Dim_Date =
ADDCOLUMNS(
    CALENDAR(DATE(2025, 1, 1), DATE(2025, 12, 31)),
    "Year", YEAR([Date]),
    "Month Number", MONTH([Date]),
    "Month", FORMAT([Date], "mmm"),
    "Quarter", "Q" & ROUNDUP(MONTH([Date]) / 3, 0)
)
```

- Open Table view and select Dim_Date. Check it has the columns Date, Year, Month Number, Month and Quarter, and 365 rows.
- Select the Date column and make sure its type is Date.
- Month names sort alphabetically (Apr, Aug, Dec …) by default. Fix this: click the Month column header → Column tools → Sort by column → Month Number. [or right click on month number and select sort ascending]
- Select the table Dim_Date in the Data pane → (right click)Table tools → Mark as date table → choose Date as the date column → OK. This tells Power BI this is the official calendar.

2. Create relationships

- Click the Model view icon (third icon on the left). You will see four boxes: Sales, Dim_Product, Dim_Region and Dim_Date. Drag them so that Sales is in the middle and the three dimension tables surround it.
- Check for relationships that Power BI created automatically (lines between tables). Double-click any existing line to inspect it. Delete incorrect or duplicate lines (right-click → Delete).
- Create the first relationship: drag the Product field from Dim_Product and drop it onto the Product field in Sales.
- A window opens. Check: Cardinality = One to many, Cross-filter direction = Single, Make this relationship active ticked. The “one” side must be Dim_Product. Click Save.
- Repeat: drag Dim_Region[Region] onto Sales[Region].
- Repeat: drag Dim_Date[Date] onto Sales[OrderDate].

Take screenshot of your model view

3. Hide technical columns

Because dimension tables hold the descriptive fields, hide the duplicate columns in Sales so that nobody picks the wrong one in a visual.

- In Model view, right-click Sales[Product] → Hide in report view.
- Do the same for Sales[Region], Sales[Category] and Sales[OrderDate].
- From now on, use Dim_Product[Category], Dim_Product[Product], Dim_Region[Region] and Dim_Date[...] in your visuals. (Customer Segment and Payment Method stay in Sales.)

Task 5: Creating visualization

In this task you build basic visuals on Page 1. You will drag raw fields into visuals; Power BI automatically applies an aggregation (a summary calculation such as Sum, Average, Count) to numeric fields.

General method for every visual

- Click an empty area of the canvas (otherwise you will change the currently selected visual instead of creating a new one).
- In the Visualizations pane, click the icon of the visual type (hover to see its name).
- Drag fields from the Data pane into the field wells (or tick the field boxes, then drag between wells if needed).
- Click the paintbrush (Format visual) icon to change titles, labels and colours.
- Resize by dragging the corner handles; move by dragging the top edge.
- First rename the page: right-click the tab Page 1 at the bottom → Rename page → Retail Sales Dashboard.

1. Sales by Category (Clustered column chart)

A **column chart** compares a number across categories using vertical bars.

Setting	Value
Visual	Clustered column chart
X-axis (categories)	Dim_Product[Category]
Y-axis (values)	Sales[Sales], aggregation Sum
Sort	Click … (More options) at the visual's top-right → Sort axis → Sum of Sales; then … → Sort descending
Format	Visual → Data labels: On; General → Title Total Sales by Category;

What to observe: which bar is tallest and which is shortest? How large is the gap?

2. Sales Over Time (Line chart)

A line chart is the best choice for showing change over time because the eye can follow the trend.

Setting	Value
Visual	Line chart
X-axis	Dim_Date[Month] (not the Date field: using Month gives one point per month)
Y-axis	Sales[Sales], aggregation Sum
Sort	… → Sort axis → Month Number → Sort ascending so that months run Jan → Dec
Format	Data labels: On; Title Monthly Sales Trend; Y-axis title off.

What to observe: the overall direction of the line, the highest month, the lowest month, and whether sales in the second half of the year differ from the first half.

3. Sales by Region (Clustered bar chart)

A bar chart is a column chart turned on its side. It is useful when category names are long or when you want to rank items.

Setting	Value
Visual	Clustered bar chart
Y-axis (categories)	Dim_Region[Region]
X-axis (values)	Sales[Sales], aggregation Sum
Sort	Sort axis by Sum of Sales, descending
Format	Data labels On; Title Total Sales by Region; one colour.

What to observe: how much larger is the top region than the others? Do the other three regions look similar?

4. Profit by Category (Clustered column chart)

Setting	Value
Visual	Clustered column chart
X-axis	Dim_Product[Category]
Y-axis	Sales[Profit], aggregation Sum
Sort	Sort axis by Sum of Profit, descending

Format	Data labels On; Title Total Profit by Category; use a different colour from Visualization 2 so profit and sales are easy to tell apart.

What to observe: compare the order of the categories here with the order in Visualization 2. Are they the same?

#	Question	Your answer
4.1	What is the total sales value shown on your card?	
4.2	Which category has the highest sales? Which has the lowest?	
4.3	Describe the shape of the monthly sales line in one or two sentences.	
4.4	Which region has the highest sales?	
4.5	Is the category with the highest sales also the category with the highest profit? What does this suggest?	